

1- Fundamentals of Computer Aided Design

2- Introduction to Computer Graphic Systems

(source book: Computer and Geometric Modeling for Engineers, Vera B. Anand, John Wiley & Sons, 1993)

Computer graphics can be thought of as a collection of computer hardware and application programs directed to one specific goal – the creation of pictures.

Computer Graphics Hardware can be classified into four categories :

- Input (*keyboard, mouse, joystick, lightpen, microphone, scanner, camera,...*)
- Storage (*magnetic disk, optic disc, flash disc, memory, paper,...*)
- Transmission (*ATAPI, IDE, RS232, COM, LPT, PCMCIA, ethernet, modem,...*)
- Output (*pin printer, laser printer, inkjet printer, pen plotter, inkjet plotter, ...*)

The graphical information is first transferred to computer system by **input** devices.

It is then **stored** internally in specific formats that depend on the device type used and is finally **transmitted** to the appropriate **output** device for viewing.

Display Technology

The **cost**, **resolution**, **color**, and **interactivity** capabilities must be considered when choosing a display device.

Resolution refers to the ability of the display to show details. Current displays reach a resolution of about 2000 X 1600, where the numbers refer to separately distinguishable spots on the screen.

The total number of colors that can be displayed at any time can vary from 8 to 4096 simultaneous colors from a total available of up to 16 million.

Today advanced graphics accelerators perform the complex three-dimensional graphics functions at the hardware level.

Figure 2.1. shows a classification chart of graphics display devices.

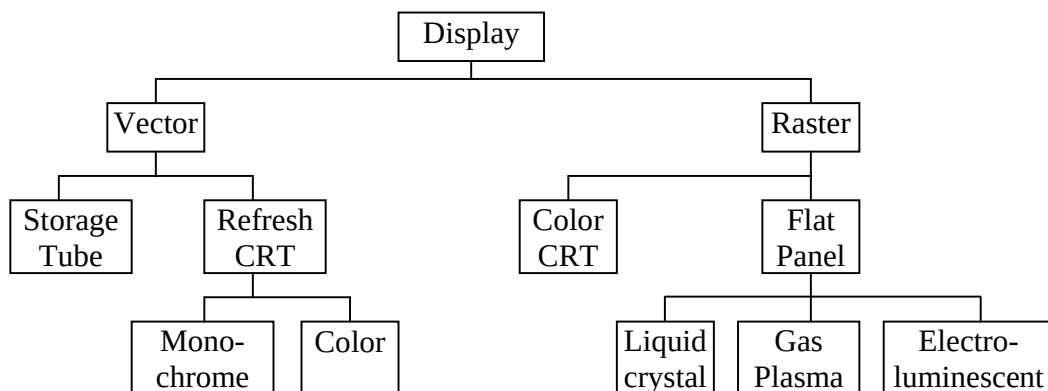


Figure 2.1. Technologies used for graphics display devices.

- Cathode Ray Tube (CRT) (*need refreshing*)
- Direct View Storage Tube (DVST) (*does not need refreshing and up to 4096x4096 res.*)
- Vector Refresh Display
- Raster Scan Display
- Color Raster Display

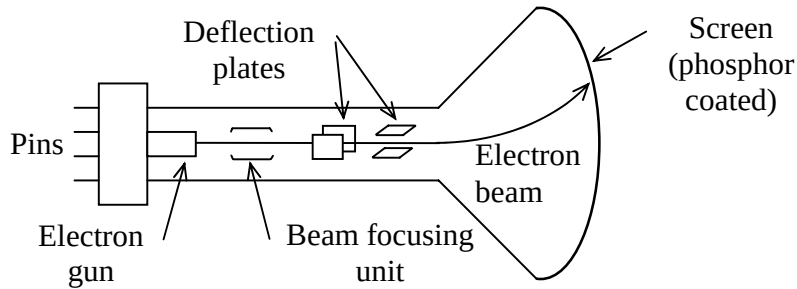


Figure 2.2. Cathode Ray Tube (CRT).

The screen of the CRT is coated with a phosphorous material that is agitated when hit by the electron beam, causing the spot to lighten. This phosphorous material is made of a mixture of various substances such as sulphur, silicon, and silver. The light emitted by the phosphor fades very quickly as the beam moves to another location.

To give the viewer the appearance of a continuous image, the electron beam must repeatedly redraw (refresh) the picture on the screen over the same spots.

For display types that are expected to store information on the screen, long-persistence phosphor is used, typically Green/White color phosphor P7 which has persistence 300 μ sec. On the other hand, if a quick update of the display is needed, a low-persistence phosphor is used, such as Green color phosphor P31 which has persistence 40 μ sec.

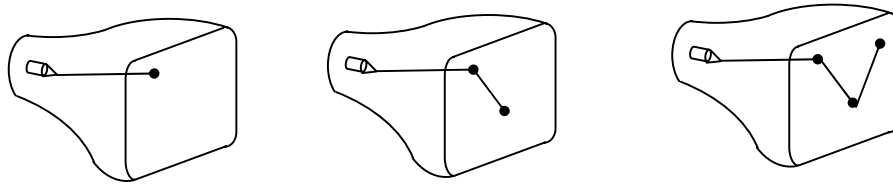


Figure 2.3. Vector Refresh Display (random scan).

The vector refresh display generates a picture by directing a beam of electrons to random points on the screen, and connecting them in vector form, as shown in Figure 2.3. The phosphor used has low persistence, and the screen has to be refreshed many times per second to avoid flickering.

In addition to the CRT, the vector refresh system requires a display buffer and a display controller. The display buffer stores the information required to create the picture, and the controller sends this information to the vector generator which transforms it into corresponding positions of the beam on the screen.

Vector displays produce a very bright and clear picture and have a high resolution. However, not only is their cost high, but they do not have the capability of color.

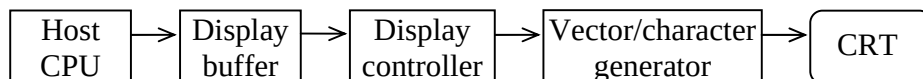


Figure 2.4. Vector Refresh System.

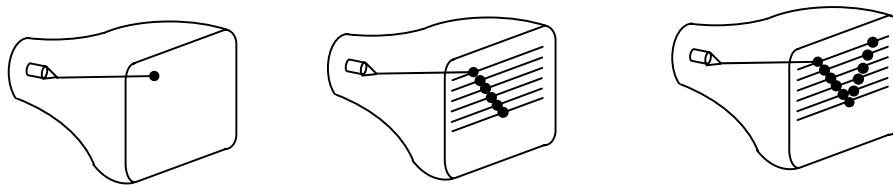


Figure 2.5. Raster Scan Display.

Raster display is a point plotting device. This type of computer terminal evolved from inexpensive television technology. The screen of raster scan display is divided into an XY matrix of dots, referred to as picture elements or *pixels*.

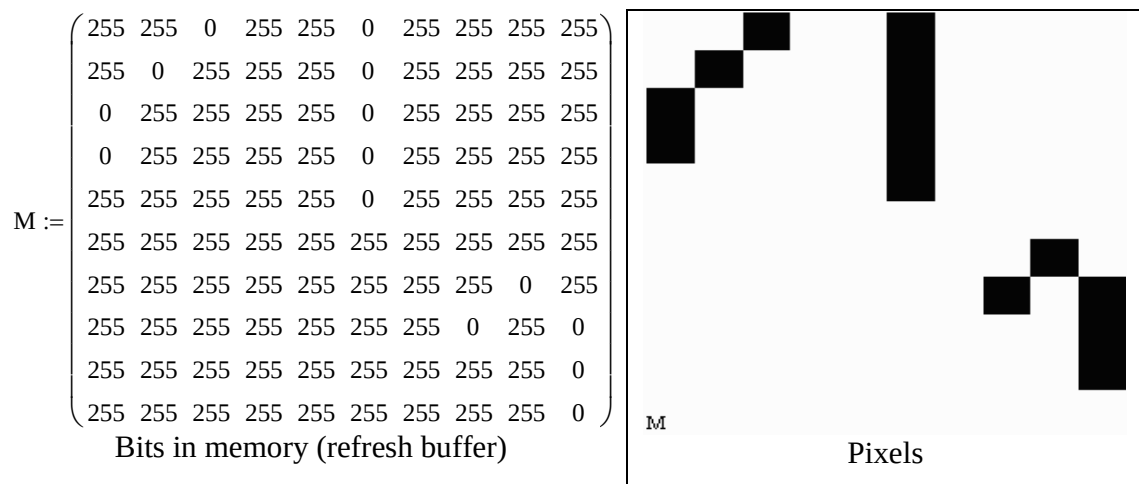


Figure 2.6. The composition of the frame buffer.

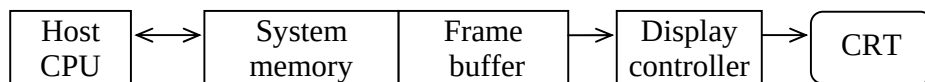


Figure 2.7. A simple raster display system.



Figure 2.8. Raster display system with dedicated display processor and memory.

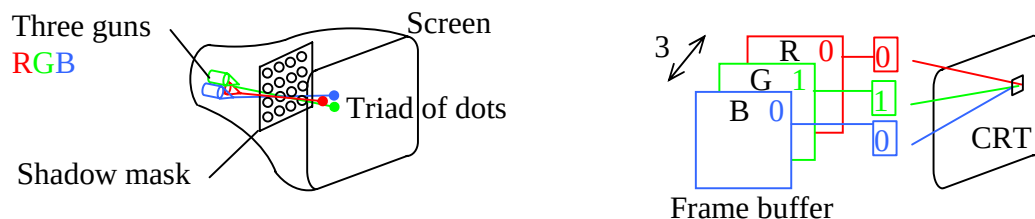


Figure 2.9. Use of shadow mask in color raster displays.

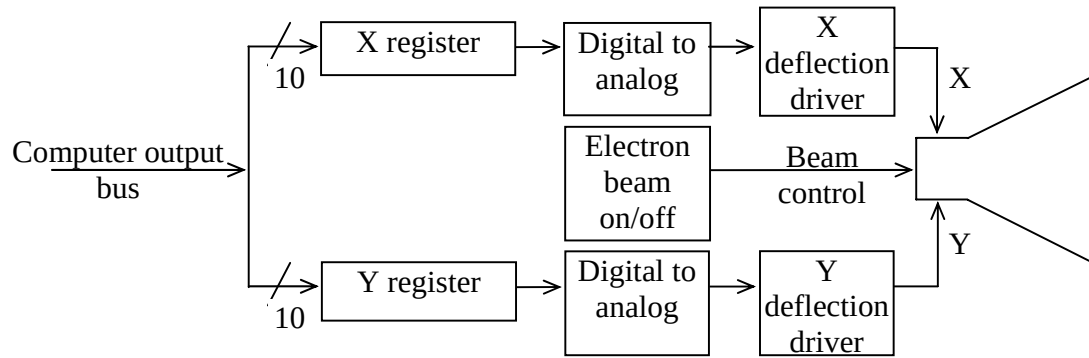


Figure 2.10. Point-plotting display.

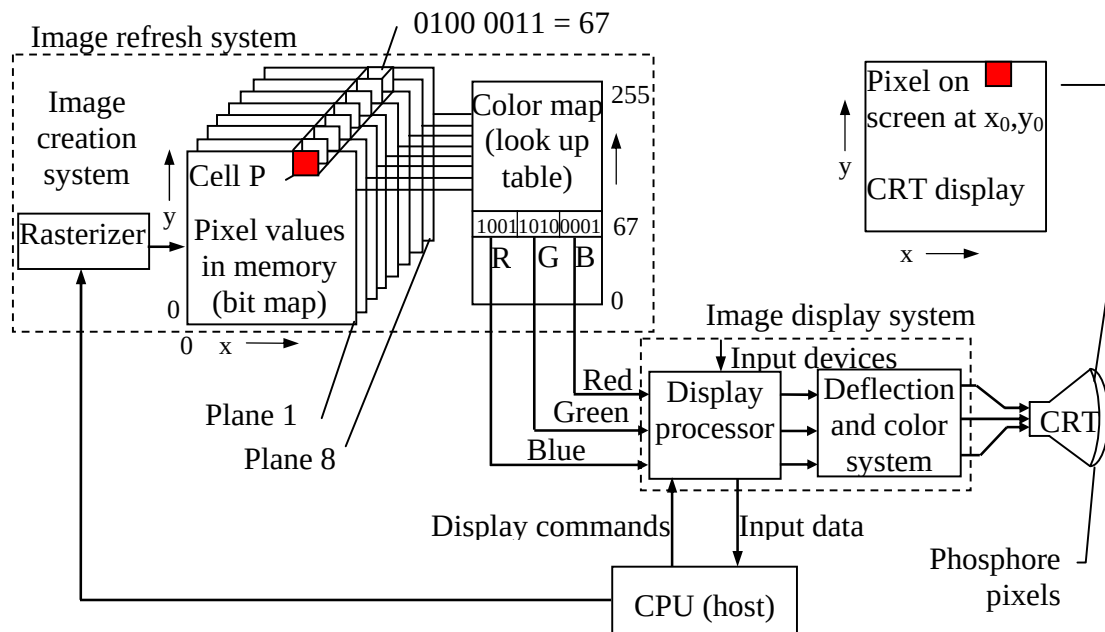


Figure 2.11. A video look-up table. A pixel with value 67 is shown.

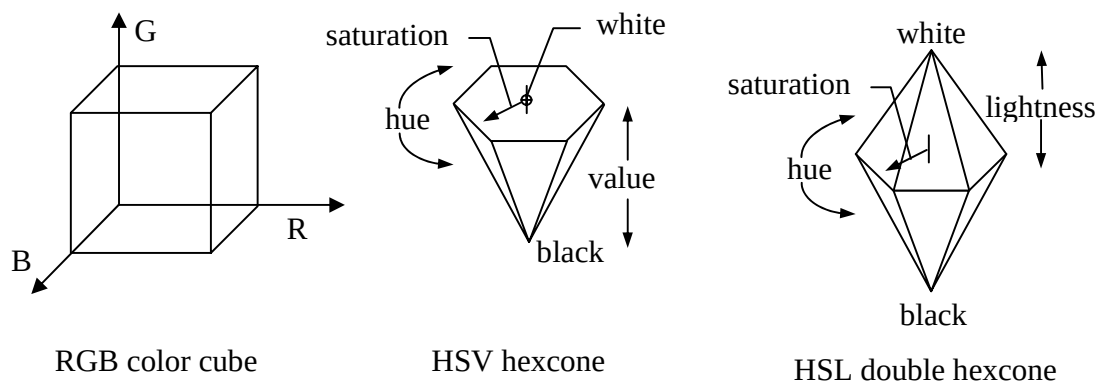


Figure 2.12. A summary of color representation.

Hardcopy Technology

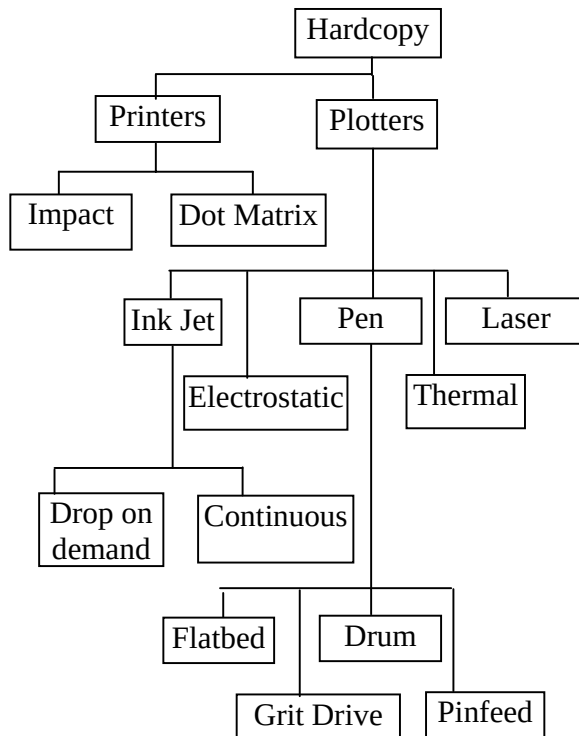


Figure 2.13. Technologies used for graphics hardcopy devices.

Graphics Standards

Several graphics standards have been developed over the years, including CORE (1977-1979), GKS (Graphical Kernel System, 1984-1985), GKS-3D (Added three-dimensional capabilities), PHIGS (Programmer's Hierarchical Graphics System, 1984), PHIGS+ include more powerful three-dimensional graphics functions, X-Windows system (1987), and OpenGL graphics standard is adapted from Unix system.

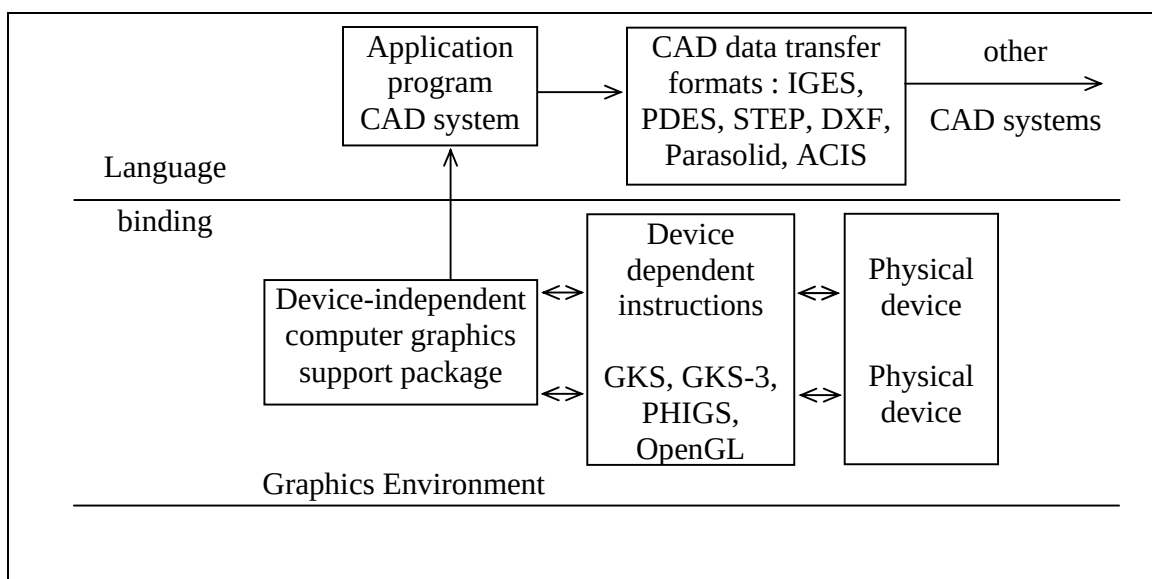


Figure 2.14. Model for standardization of the graphics environment.